Standard Designs

RaBiT

SMARTIES

SMARTIES Documentation

SMARTIES is a two phase adaptive design with an internal pilot study for feasibility assessment and unblindedc sample size re-estimation (Phase I), followed by a main trial for testing efficacy with adjusted sample size when needed (Phase II).

Quick Start

1. **Step 1 (Feasibility):** Enter p_0, p_1, α, target power, and number of candidate designs. Choose Auto/Manual and click `Use this design` to lock the internal pilot design.
2. **Step 2 (Initial main trial sample size calculation):** Enter α, power, SD, and planned mean difference d. Click `Lock original main trial sample size` to lock N_{orig}.
3. **Step 3 (SSR):** Enter d_{obs}, γ_l, γ_u, nsims, seed, then click `Run Sample Size Re-estimation`.
4. **Report:** Review the final summary and click `Download Word report`.

Step 1: Internal Pilot (Feasibility)

Goal: decide whether the trial is feasible to proceed based on a one-stage exact test for a binary feasibility endpoint.

- **p_0**: unacceptable feasibility success rate.
- **p_1**: acceptable feasibility success rate (must satisfy $p_1 > p_0$).
- **α** (one-sided): Type I error for feasibility test.
- **Power**: probability of declaring feasibility under p_1.
- **Output**: interim total sample size $\mathbf{n_{int}}$ and feasibility cutoff $\mathbf{r_{feas}}$.

Step 2: Initial Main Trial (Efficacy)

Goal: compute the initial planned sample size for a two-arm superiority test under user-specified α, power, SD, and planned mean difference d.

- **Output**: per-arm $\mathbf{n_{orig}}$ and total $\mathbf{N_{orig}}$.
- **Tip**: ensure per-arm interim n (from Step 1) is smaller than per-arm planned n (from Step 2) so the interim information fraction is valid.

Step 3: SSR using the CHW conditional power rule

Goal: re-estimate the required sample size based on conditional power computed from interim data.

Key quantities

- $\mathbf{d_{obs}}$: observed interim mean difference (user input).
- **t**: interim information fraction (approximately $n_{int}/2$ divided by n_{orig}).
- $\mathbf{CP_{plan}}$: conditional power under planned effect size.
- $\mathbf{CP_{obs}}$: conditional power under observed interim effect size.
- $\boldsymbol{\gamma_l}$, $\boldsymbol{\gamma_u}$: lower/upper thresholds defining zones.

Decision zones (CHW)

- **Futility:** $CP_{obs} < \gamma_l \times CP_{plan}$ (keep original N; consider stopping).
- **Promising:** $\gamma_l \times CP_{plan} \leq CP_{obs} \leq \gamma_u \times CP_{plan}$ (apply CHW inflation / top-up).
- **Success:** $CP_{obs} > \gamma_u \times CP_{plan}$ (keep original N).

The app summarizes SSR via simulation (nsims runs), producing the distribution of suggested N and the zone proportions.

How to read the outputs

- **Design header**: locked Step 1/2 inputs, final α, and interim t.
- **Key metrics table**: effect sizes, CP metrics, and suggested sample size summary.
- **Decision rule & interpretation**: CHW rule + dominant zone statement.
- **Plots**: zone counts + histogram of suggested total N.
- **Word report**: timestamp + summary + rule + schema figure.